RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Sunday, 16 August 2026, 10:37 PM
State	Finished
Completed on	Sunday, 16 August 2026, 10:38 PM
Time taken	36 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int found = 0;
int elem1, elem2;

// Checks if any pair with one element in [low, mid] and one in [mid+1, high] sums to x
void checkCrossing(int arr[], int low, int mid, int high, int x) {
    int i = mid;
    int j = mid + 1;

    while (i >= low && j <= high) {
        int sum = arr[i] + arr[j];
        if (sum == x) {
            elem1 = arr[i];
            elem2 = arr[j];
            found = 1;
            return;
        } else if (sum < x) {
            j++;        // increase sum
        } else {
            i--;        // decrease sum
        }
    }
}

void twoSum(int arr[], int low, int high, int x) {
    if (found || low >= high) {
        return;
    }

    int mid = low + (high - low) / 2;

    // Divide: solve left half
    twoSum(arr, low, mid, x);
    if (found) return;

    // Divide: solve right half
    twoSum(arr, mid + 1, high, x);
    if (found) return;

    // Combine: check pair crossing the two halves
    checkCrossing(arr, low, mid, high, x);
}

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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